

The Elements of DeFi

DeFi does not have an innovation problem. It has a vocabulary problem.

Nearly every protocol ever shipped is a recombination of about fifty recurring mechanisms — and once you can name them, most of the industry stops looking novel and starts looking legible.

Call it a periodic table if you like, but it is not one: no periodic law, no atomic number, nothing recurring down a column. Chemistry discovered its elements. These were designed, in a hurry, by people shipping to production — which is exactly why the combinations are the part worth writing down.

THEME

Dark Light

v1.0 · reviewed 2026-08-04

48 core · 10 candidate · 10 contested · 29 laws · 19 hazards

READ A PROTOCOL

Pick one. It lights up in the table below, numbered in the order each part became necessary. Four of these are dead — and those are the instructive ones.

Aave v3 Pooled lending 10 elements	Uniswap v3 Spot exchange 4 elements	Maker / Sky CDP stablecoin 10 elements	Liquity v1 CDP stablecoin 6 elements
GMX v2 Perpetual futures 8 elements	Lido v2 Liquid staking 5 elements	CoW Protocol Intent exchange 4 elements	CCTP v2 Fast Cross-domain transfer 4 elements
Centrifuge Real-world credit 8 elements	Terra / Anchor + Algorithmic stablecoin 6 elements	Mango Markets + Perpetual futures 6 elements	Euler v1 + Pooled lending 7 elements

Rows are how much must already exist

S0 needs nothing but the ledger. S4 needs other people to agree. Deeper is not more dangerous — it is more dependent, and confusing those two is how you misprice a protocol.

Colour only ever means hazard

There is exactly one coloured thing in this interface and it is danger. Everything else earns its distinction from position, shape and lightness.

Blocks are families of substitutes

Everything in a block answers one question. Swap Cp for Cl and you still have a DEX. That is what a block means and the only thing it means.

The shape is the finding

Families cluster at one depth — credit is all S3, pricing all S1. And S3 is enormous: most of DeFi is machinery for owing, pricing and unwinding obligations.

VIEW

The table

Laws

Hazards

ARRANGEMENT

By depth

By family

DIMENSION

Flat

3D

READING ORDER

By group

By stratum

By ID

SEARCH

symbol, name or ID

Row = prerequisite depth, exactly. Blocks inside a row are families of substitutes. Order within a block is not semantic.

E001 S0 Sh Pro-rata share accounting ■ core	E002 S0 Ix Index-based accrual ■ core	E003 S0 Rb Rebasing accounting ■ core	CSM S1 CSM Constant sum ✓ $\frac{n}{\dots}$ degenerate limit: not an element ■ core	E004 S1 Cp Constant-product invariant ■ core
E005 S1 Wg Weighted-geometric... ■ core	E006 S1 St Stable-hybrid invariant ■ core	E007 S1 Cl Concentrated liquidity ■ core	E008 S1 Pm Oracle-priced inventory curve ■ core	E009 S1 Ob On-chain order book ■ core
E010 S1 Rf Request for quote ■ core	E011 S2 Ba Batch-auction clearing ■ core	E012 S4 In Intent & solver execution ■ core	E039 S1 Ag Aggregation & routing ■ core	E040 S1 Fl Atomic flash liquidity ■ core
E013 S3 Pl Pooled lending ■ core	E014 S3 Im Isolated lending market ■ core	E015 S3 Cd Collateralized-debt minting ■ core	E016 S3 Uc Undercollateralized credit ■ core	E017 S3 Ft Fixed-term debt ■ core
E018 S3 Ct Collateral-threshold test ■ core	E019 S3 Li Incentivized liquidation ■ core	E020 S3 Ad Auto-deleveraging ■ core	E021 S3 Sl Socialized-loss allocation ■ core	E022 S3 Bs Staked backstop ■ core

E023S3 Pf Perpetual funding transfer core	E024S3 Op Option payoff core	E025S3 Tr Tranche waterfall core	E026S3 Cv Mutual cover pool core	E027S3 Py Principal/yield separation core
E057S3 Sv Servicing & determination... n candidate: recurrence evidence short	E060S3 Dp Directional position & hedge... n candidate: recurrence evidence short	E028S2 Ex External data oracle core	E029S2 Tp Time-weighted price core	E030S2 Oa Optimistic assertion oracle core
E031S2 At Reserve / NAV attestation core	E032S2 Sr Streaming accrual core	E033S2 Ep Epoch-gated transition core	E034S2 Wq Withdrawal queue core	E035S2 Em Protocol-funded emissions core
E058S2 Fd Surplus & fee distribution n candidate: recurrence evidence short	E051S3 Gs Sponsored-fee liability n candidate: recurrence evidence short	E036S4 Tg Delayed-governance... core	E037S4 Up Mutable implementation... core	E038S4 Gp Guardian or pause core
E050S4 Au Delegated execution scope core	E041S4 Xm Cross-domain message... core	E042S4 Xf Cross-domain asset transfer core	E053S4 RL Resource lock / reservation n candidate: recurrence evidence short	E054S4 Of Optimistic fill & reimbursement n candidate: recurrence evidence short
E043S3 Rd Direct redemption right core	E044S3 Ps Peg-swap module core	E045S3 As Algorithmic supply adjustment core	E046S2 Aw Permission / identity gate core	E047S2 Sb Shielded-balance state core
E048S2 Sd Selective-disclosure proof n candidate: recurrence evidence short	E056S2 Fz Freeze / forced transfer n candidate: recurrence evidence short	E049S4 Rs Restaking / shared security n candidate: recurrence evidence short	E059S3 VL Staking & validator... n candidate: recurrence evidence short	

S0

Ledger-local claims3 elements

G01

Claims & accounting

3

S1

Deterministic transformation10 elements

G02

Pool pricing

6

G03

Execution

2

G04

Liquidity catalysis

2

S2

Measured or time-conditioned14 elements

G03	Execution	1
G08	Truth	4
G09	Time & queueing	3
G10	Incentives	2
G14	Access & privacy	4

S3

Obligation & solvency22 elements

G05	Credit	5
G06	Solvency	5
G07	Risk transfer	7
G11	Control & authority	1
G13	Stability	3
G16	Staking	1

S4

Multi-agent & cross-domain10 elements

G03	Execution	1
G11	Control & authority	4
G12	Cross-domain	4
G15	Security reuse	1

Contested register – 10 entries

Held outside the table because a necessary criterion is genuinely disputed.
Not usable in a formula without a note. The separation means “argued about”, not “filed away”.

P001 Bc Bonding-curve issuance n contested: note required	P002 Tw Time-weighted AMM execution n contested: note required	P003 Da Dutch-auction descent n contested: note required
P004 Cg Credit delegation n contested: note required	P005 Ir Insurance reserve fund n contested: note required	P006 Zk Verifiable state proof n contested: note required
P007 Ve Vote-escrow allocation n contested: note required	P008 Kg Credential-gated transfer n contested: note required	P009 Ua Unified-balance ledger n contested: note required



HOW TO THINK WITH IT

Six questions, in this order. Run them against anything — a protocol you are auditing, a pitch deck, your own design — and the table stops being a diagram and becomes a method. Question three pays for the other five.

1

What are its parts?

Name every recurring financial state transition and refuse everything else. ERC-4626 is a socket, not a mechanism. A hook is somewhere to put an element, not an element. “Points” is marketing. Most whitepapers describe three real mechanisms and forty pages of scenery.

2

What did each part force?

Parts are not chosen freely — each one drags in the next. Leverage forces a truth source, a solvency test and a way for the position to end. A protocol with leverage and only two of those three is not innovating. It is missing a part, and you have just found it.

3

Where does it read the outside world?

This is the question that pays. Every catastrophic loss this table explains runs through the price, not the code. Ask what it would cost to move that input — measured against pool depth, never against the attacker’s balance — and if the answer is less than what can be borrowed against it, the protocol is already lost.

4

Does anything feed itself?

Reflexive collateral is the most reliable way to lose everything ever deployed on-chain. A token that is simultaneously the collateral, the market that prices it, and the capital backing it has three defences that all fail on the same news. This is the one shape that goes to zero rather than merely down.

5

Who eats the loss?

For every fee, delay, slippage and advance, name the claim class or the underwriter. An unassigned loss has not vanished — somebody is carrying it without being paid to, and usually it is the depositor who never read this far. If the documentation cannot name them, that is the answer.

6

What did the formula fail to say?

The residue, and it is the most useful line on the page. For a lending market it is small. For anything touching real-world assets it is larger than the formula and contains the actual risk. A decomposition that reports no residue was not finished.

LEGEND

Stratum S0 → S4



Prerequisite depth, not risk. Carried by the band,
the S-token and lightness — never hue.

 core

 candidate: recurrence evidence
short

contested: note required

 hazard: forbidden

 hazard: elevated

 hazard: unverifiable



Hazard membership is categorical. It is never a probability, because no exposure or survivor counts exist.

／ degenerate limit: not an element

$\frac{n}{\text{counted}}$ no denominator – failures counted, survivors never were

–i→ Interface – Can these exchange the required calls, tokens, messages, proofs?

–e→ Economic – Do combined incentives, liquidity and loss allocation stay solvent?

–t→ Trust – Who can attest, censor, upgrade, pause, mint, freeze?

–n→ Informational – Who observes the order before it settles?

This table explains roughly two thirds of DeFi incidents by count and a minority of them by dollars. That is not a defect in the table — it is the most useful finding on this page. Your largest risk is not your financial design. It is your keys, your signing interface and your build pipeline. Get the mechanism right and you have eliminated the failures people write papers about, not the ones that actually took the money.